

Biometry

Today's agenda:

Introductions,

Syllabus,

Why we need stats,

Reproducibility and Openness

Introductions

Please take a moment to tell us about yourself:

01

Your Name

Introduce yourself to the class, including what you prefer to be called and your pronouns if you'd like to share.

02

Research

Describe your current research interests, what systems or questions you study, or what you hope to work on.

03

Fun Fact

Tell us something interesting or unique about yourself—hobbies, favorite travel spots, pets, or hidden talents!

Syllabus: basic stuff

Class Meeting Days: T, Th

Class Meeting Time: 2:00 – 3:50 pm

Class Meeting Location: DAV 265

Instructor: Dr. Brian Maitner

Office Location: DAV 226 (but I'm usually in URL 106, because it's warmer)

Office Hours: TBD - ?

Email: bmaitner@usf.edu

Up to date syllabus: https://github.com/bmaitner/biometry_course

Course Github page

 bmaintr time and location update b495611 · 50 minutes ago  117 Commits		
 R_scripts	updated urls	19 hours ago
 assignments	changing pipe to base R	yesterday
 data	removed larger marine organism data files, updated readm...	19 hours ago
 example_R_commented	changing pipe to base R	yesterday
 example_R_markdown	changing pipe to base R	yesterday
 final	final	9 months ago
 lectures	notes on pending lecture updates	19 hours ago
 midterm	changing pipe to base R	yesterday
 quizzes	q6	8 months ago
 renv	renv	9 months ago
 syllabus	time and location update	50 minutes ago
 .Rprofile	renv	9 months ago
 .gitignore	gi update, removed old syllabus, switching to md file that re...	yesterday
 LICENSE	Initial commit	between

Full of useful stuff

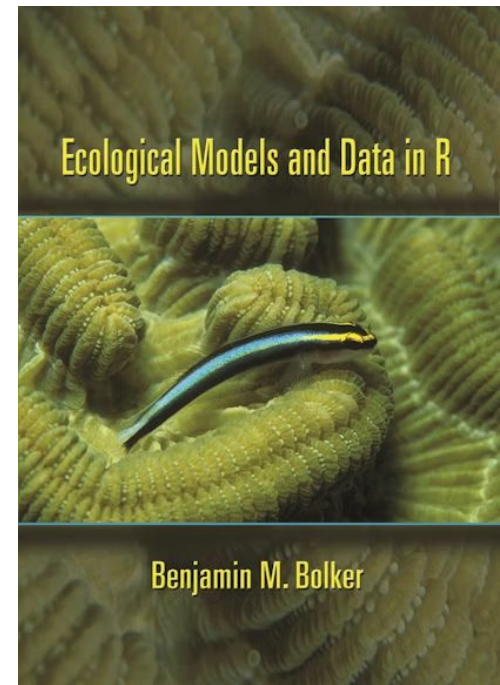
- Syllabus
- Lectures
- Previous quizzes

Syllabus: Course structure

Flipped course: read before class, work during class

Assignments will focus on data of your choosing

Graded Items	Percent of Final Grade
In-class participation	20%
In-class quizzes	20%
Assignments (4x)	20%
Midterm (take-home)	20%
Final (presentation + take-home)	20%



Digital copies available
through the library for free

Course Expectations



Participate!

Ask questions. Share ideas, experiences, and knowledge!



Be Respectful.

Create a welcoming environment for everyone to learn, collaborate, and grow together.

Recommendations



**Read the
syllabus**



Read the book



**Take notes on
paper**



Plan ahead



Bring a laptop

Questions so far?

What will you get out of this class?

In general:

- Ability to do common analyses and visualizations
- Ability to go further on your own

What will you get out of this class?

More specifically:

Student Learning Outcomes

- Load data into R and conduct common data-wrangling tasks
- Create data visualizations using base R and ggplot2
- List the common types of statistical distribution and provide examples of when they might apply.
- Choose appropriate analyses for given ecological questions and datasets.
- Demonstrate an ability to troubleshoot and de-bug R code.
- Explain how they would approach a novel coding problem
- Apply the skills learned to their own work

Learning Stats

People learn differently and stats can be non-intuitive



THE BOOK

Exposes you to verbal and mathematical descriptions of key statistical concepts.



IN CLASS

Hands-on coding and interactive visualizations help build intuitive understanding.



THE GOAL

Providing multiple teaching methods ensures that at least one appeals to everyone.

Learning Stats



What to Avoid

**Don't focus on
memorization**



What to Focus On

**DO focus on practice and
understanding**

A note on AI

- We'll cover AI later in the semester
- Strongly recommend that you start out without it
- You need reasonable fundamentals to be able to effectively use AI

Questions so far?

Why do we need statistics?

Why do we need statistics?

01 . COGNITIVE LIMITS

Imperfect Minds

We must overcome human biases, logical fallacies, and cognitive limitations.

02 . MEASUREMENT

Effect Sizes

Accurately quantifying the strength and magnitude of different relationships.

03 . VALIDATION

Hypothesis Testing

Systematically validating assumptions against empirical data.

04 . FORECASTING

Making Predictions

Building robust models to anticipate future outcomes and trends.

05 . CLARITY

Visualizations

Making complex numerical relationships instantly clear and intuitive.

06 . ETC

Lots of Other Stuff

From optimizing systems to driving data-informed strategic decisions.

What is R and why is it useful?

THE CORE IDEA

Scripted Language

Type **commands** instead of clicking buttons. This fundamental shift unlocks precision and power in data analysis.

REPRODUCIBILITY

Easily Re-run & Revise Analyses

Modify your inputs or parameters and re-execute entire workflows instantly.

EFFICIENCY

Re-use Code Across Projects

Build on top of previous work by recycling robust scripts for new datasets.

DOCUMENTATION

Code as a Methods Section

Your codebase serves as a transparent, step-by-step record of your methodology.

What is R and why is it useful?

01 . COST-EFFECTIVE

R is Free to Use

Anyone can download and use R without licensing costs, lowering the barrier to entry.

02 . EXTENSIBILITY

User-Contributed Packages

A massive ecosystem of user-developed packages is contributed for free, extending its capabilities.

03 . VERSATILITY

Incredible Flexibility

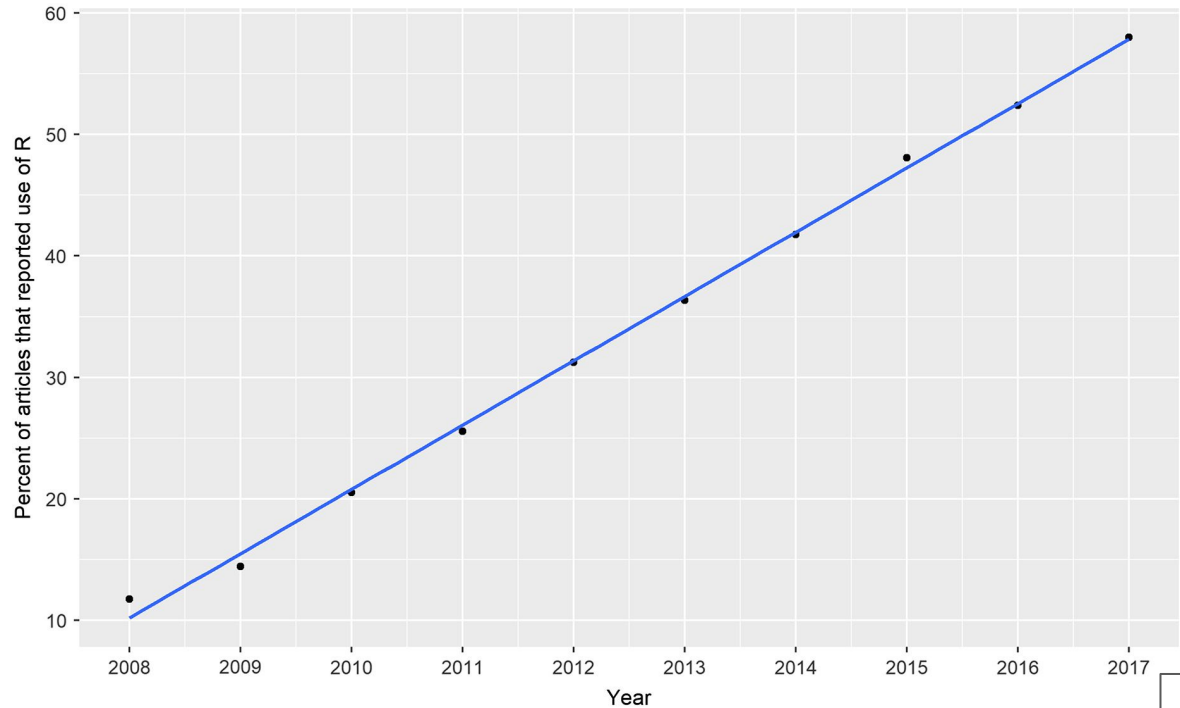
The vast range of libraries allows researchers to handle virtually any quantitative analysis task.

04 . COMMUNITY

Popular & Supported

An active, global developer and user base provides vast documentation and peer-to-peer help.

What is R and why is it useful?



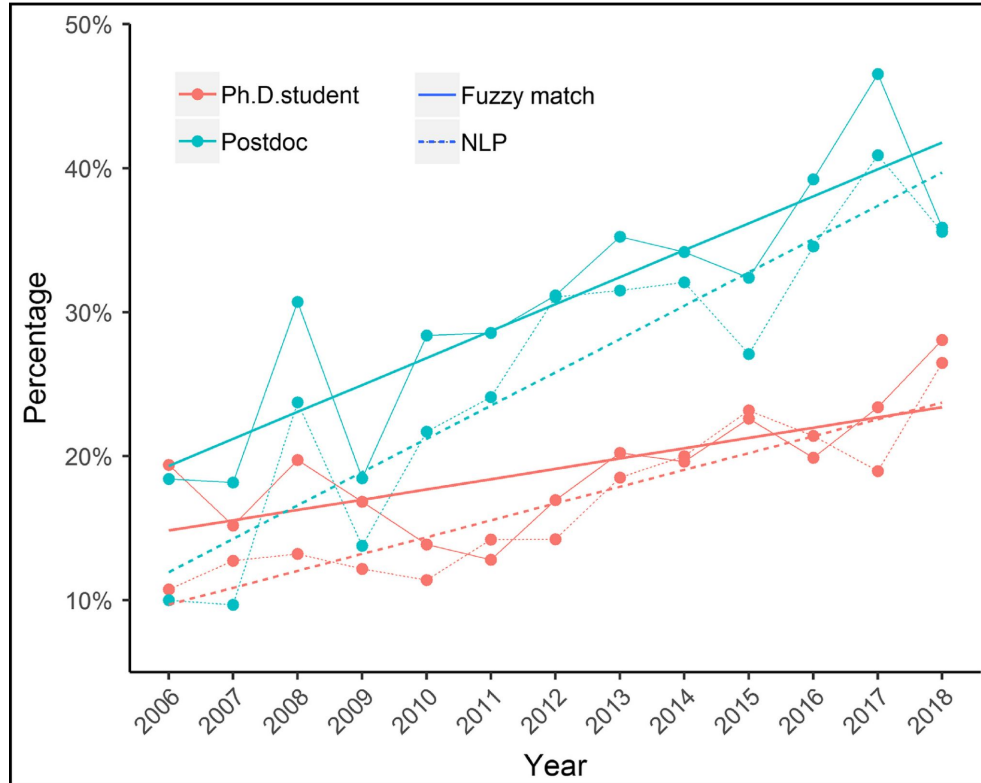
esa

ECOSPHERE

Evaluating the popularity of R in ecology

JIANGSHAN LAI^{1,2}, CHRISTOPHER J. LORTIE^{3,4}, ROBERT A. MUENCHEN⁵, JIAN YANG⁶ AND KEPING MA¹

What is R and why is it useful?



Transparency & Reproducibility

Science needs to be open, clear, and verifiable.



Transparency

What was done is clear



Reproducibility

Can be re-done with the same results



R helps with both of these!

Transparency and Reproducibility

2 MATERIALS AND METHODS

2.1 Data collection

2.1.1 List of ecology and evolution publications citing R

To generate a list of papers in ecology and evolution that likely made use of the R programming language (R Core Team, 2023), we performed a query on the Scopus database (<https://www.scopus.com>) using the `scopus` R package (Muschelli, 2019). We searched Scopus (performed August 19, 2023) for peer-reviewed journal articles that: (1) included the words "ecology" or "evolution" in an "all fields" search (which searches article titles, keywords, abstracts, and journal titles); (2) were published in journals within the subject area "agriculture and biological sciences"; (3) were published after January 1, 2010; (4) were written in English (as this is currently the dominant language of publication in ecology and evolution; Mauranen et al., 2010); and (5) included a citation of R in their reference list.

2.1.2 Checking for code and data availability

We manually evaluated a randomly chosen subset of the publications on our overall list. We selected a total of 1001 papers, evenly distributed across the time period (77 year * 13 years). Papers that cited R but did not use it (or were unclear on whether they used it; $n = 3$) were discarded and replaced by a randomly selected paper from the same year. For each publication in this subset, we manually identified whether the publication shared any R code, either as supplementary information, or via a link (e.g., to a GitHub repository). For each paper, we (i) checked for the presence of code in supplemental material, (ii) skimmed publications for code and data availability statements, (iii) searched through publications for terms associated with code (i.e., "code", "supplement", "appendix", "R", "script", "GitHub"), and (iv) searched publications for URLs. Papers were scored with a binary variable indicating whether they shared R code or not. We did not distinguish between publications which shared sufficient code for reproduction and those which did not. We also did not attempt to return the code or assess its reproducibility, and only recorded the presence of any code, even if it was incomplete. Where code was included, we recorded the license the code was provided under, or lack thereof. We also assessed whether publications were open access and whether they shared open data in order to understand the importance of open code relative to these other open-access components. Open access information was provided by the `scopus` R package (Muschelli, 2019). Open data was scored as a binary variable indicating whether the authors shared the full set of raw data underlying the analyses or not. To control for differences in citation rates among journals, we downloaded impact factor information using the `scholar` R package (Körner, 2016) on June 16, 2023. To estimate the proportion of publications which use R but do not properly cite it, we screened 130 randomly selected publications evenly distributed across the time period. These publications were selected using identical criteria to the publications that did cite R, except that they did not include R in their list of references.

2.2 Checking for code citations

Where code was shared in a citable location such as a DOI or URL ($n = 33$), we assessed whether the code itself was cited by querying the Scopus database for the URL (and DOI, where appropriate) using the `scopus` R package (Muschelli, 2019). Publications where code was shared in appendices or supplementary information ($n = 22$) were excluded, as there was no way of distinguishing citations of the code with citations of the publication itself.

2.3 Analyses

All analyses were conducted in R version 4.3.0 (R Core Team, 2023). All R scripts underlying these analyses are available at: https://github.com/bmaitner/R_citations and via Zenodo (Maitner & Lei, 2024). For data processing, we used the R packages `stringr` version 0.9.12 (van der Lee, 2014), `tidyr` version 2.0.0 (Wickham et al., 2018), `ggplot2` version 2.1.1 (D'Agostino McGowan & Bryan, 2023), and `googlehttplib2` version 1.1.1 (Bryan, 2023). For analyses, the packages `binde` version 1.0.25.1 (Baker & R Development Core Team, 2023), `binde` version 1.4.4 (Zhang, 2022), `MAIn` version 1.4.7.5 (Barton, 2023), `scopus` version 0.6.6 (Muschelli, 2019), `scholar` version 2.6 (Zhang, 2018), `scholar` version 0.2.4 (Körner, 2016), and `stats` version 4.3.0 (R Core Team, 2023). For plotting, the packages `ggplot2` version 3.4.4 (Wickham, 2016), `ggmargins` version 0.5.5 (Alpha), `tidy`, and `question` version 0.7.8 to identify all R packages used.

2.3.1 Proportion of papers sharing code over time

We tested for a trend in code-sharing over time by modeling code sharing (binary, `yes/no`) as a function of the year (relative to 2010) using a generalized linear model. Modeling was performed using the function `glm` in the `stats` R package (R Core Team, 2023) with a binomial error distribution. We similarly tested for temporal trends in two other open-science components, open access publication (binary) and open data (binary). We also tested whether open access or open data papers were disproportionately likely to share code using `chi.square.test` via the `chisq.test` function in the `stats` R package (R Core Team, 2023).

2.3.2 Impact of code sharing on citations

We additionally modeled the relationship between code sharing and citation count using generalized linear models in R. We modeled the dependent variable (cumulative number of citations of each article by 2022) using a Poisson distribution, which models the number of independent events occurring within a period of time (Baker, 2008). In addition to the predictor variable for code sharing (binary, `yes/no`), we included other variables that were hypothesized to influence citation count. Data sharing (binary, `yes/no`) may increase citation counts as readers may cite papers at data sources (Christensen et al., 2019; Priesner et al., 2007). Open access (binary, `yes/no`) may also increase citation counts by reaching a broader set of readers (Fang et al., 2017). Publications accumulate citations over time, and so citation count should increase with publication age (continuous, 1–13 years). Finally, publications in higher impact journals may be more likely to be read and cited, and hence, journal impact factor (continuous) may also influence citation counts. In addition to main effects, we considered two classes of interactions: (1) interactions between publication age and other main effects, which are appropriate if a main effect modifies the rate at which a publication accumulates citations over time; and (2) interactions between open access criteria (i.e., open access, open code, and open data), which are appropriate if there are synergistic effects of meeting multiple open access criteria. We compared 11 models (including one null model that differed in complexity and that represented different hypotheses regarding the factors that influence citations (Table 1). Continuous variables were scaled and centered. Overall model pseudo R^2 for the best-performing model was calculated using the function `r.squaredGLMM` in the `mgc` package (Zhang, 2018).

TABLE 1. Candidate models of citation count.

ID	Models	#F	AIC
1	Citations ~	13	0.0
	Impact factor ~ Age ~		
	Code shared ~ Age ~		
	Open access ~ Age ~		
	Data shared ~ Age ~		
	Data shared ~ Code shared ~		
	Code shared ~ Open access ~		
	Open access ~ Data shared		
2	Citations ~	9	8343
	Impact factor ~ Age ~		
	Code shared ~ Age ~		
	Data shared ~ Age ~		
	Data shared ~ Code shared		

Code sharing in ecology and evolution increases citation rates but remains uncommon

Brian Maitner^{1,2} | Paul Efrén Santos Andrade³ | Luna Lei⁴ | Jamie Kass⁵ | Hannah L. Owens^{6,7,8} | George C. G. Barbosa⁹ | Brad Boyle¹⁰ | Matiss Castorena¹⁰ | Brian J. Enquist^{10,11} | Xiao Feng¹² | Daniel S. Park^{13,14} | Andrea Paz^{15,16} | Gonzalo Pinilla-Buitrago¹⁷ | Cory Merow¹⁸ | Adam Wilson²

Transparency and Reproducibility

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All analyses were conducted in R version 4.3.0 (R Core Team, 2023). All R scripts underlying these analyses are available at: https://github.com/bmaitner/R_citations and via Zenodo (Maitner & Lei, 2024). For data processing, we used the R packages singlr (version 0.9.12 (van der Lee, 2014), rstan version 2.20 (Wickham et al., 2019), ggplot2 (version 2.1.1 (D'Agostino McGowan & Bryan, 2023), and googlescholar (version 1.1.1 (Bryan, 2023)) for analyses, the packages binde (version 1.0.25.1 (Bulker & R Development Core Team, 2023), DfM (version 1.4.4 (Zhang, 2023), Mafin (version 1.47.5 (Baron, 2023), rscopus (version 0.6.6 (Muschelli, 2019), rscopus (version 2.6 (Zhang, 2018), scholar (version 0.2.4 (Körner et al., 2016), and rstan version 4.3.0 (R Core Team, 2023)) for plotting, the packages ggplot2 (version 3.4.4 (Wickham, 2019), ggvis (version 0.5.5 (Aghajani, 2023), and qversion (version 0.7.8 to identify all R packages used.

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We tested for a trend in code-sharing over time by modeling code sharing (binary, yes/no) as a function of the year (relative to 2010) using a generalized linear model. Modeling was performed using the function glm in the stats R package (R Core Team, 2023) with a binomial error distribution. We similarly tested for temporal trends in two other open-science components, open access publication (binary) and open data (binary). We also tested whether open access or open data papers were disproportionately likely to share code using chi-square tests via the chi2test function in the stats R package (R Core Team, 2023).

2.3.2 Impact of code sharing on citations

We additionally modeled the relationship between code sharing and citation count using generalized linear models in R. We modeled the dependent variable (cumulative number of citations of each article by 2023) using a Poisson distribution, which models the number of independent events occurring within a period of time (Bulker, 2008). In addition to the predictor variable for code sharing (binary, yes/no), we included other variables that were hypothesized to influence citation count. Data sharing (binary, yes/no) may increase citation counts as readers may cite papers as data sources (Christensen et al., 2019; Priesner et al., 2007). Open access (binary, yes/no) may also increase citation counts by reaching a broader set of readers (Ting et al., 2013). Publications accumulate citations over time, and so citation count should increase with publication age (continuous, 1–13 years). Finally, publications in higher impact journals may be more likely to be read and cited, and hence, journal impact factor (continuous, 0–13) may be positively associated with citation count. In addition to main effects, we considered two classes of interactions: (1) interactions between publication age and other main effects, which are appropriate if a main effect modifies the rate at which a publication accumulates citations over time; and (2) interactions between open science criteria (i.e., open access, open code, and open data), which are appropriate if there are synergistic effects of meeting multiple open science criteria. We compared 11 models (including one null model that differed in complexity and that represented different hypotheses regarding the factors that influence citations (Table 1). Continuous variables were scaled and centered. Overall model pseudo R^2 for the best-performing model was calculated using the function rsquaredGLMM in the r2 package (Zhang, 2018).

TABLE 1. Candidate models of citation count.

ID	Models	#F	AIC
1	Citations ~	13	8.0
	Impact factor ~ Age ~		
	Code shared ~ Age ~		
	Open access ~ Age ~		
	Data shared ~ Age ~		
	Data shared ~ Code shared ~		
	Code shared ~ Open access ~		
	Open access ~ Data shared ~		
2	Citations ~	9	834.3
	Impact factor ~ Age ~		
	Code shared ~ Age ~		
	Data shared ~ Age ~		
	Data shared ~ Code shared ~		

Transparency and Reproducibility

Note: many journals and funders are now requiring code to be published

Questions on any of this?

Next time:

Before class: read 1.1 - 1.3

During class:

- Discuss 1.1-1.3
- Installing and setting up R, RStudio, and Github (if needed)